



Linkage, Not Similarity: Context-Sensitive Retrieval for Long-Term Agent Memory

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Revision 1

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Abstract

Vector similarity retrieves what sounds related. Long-term memory has to retrieve what is connected: the turn that answers a question is usually linked to the turn that matches it by a person, an event or a date, not by wording. We measure this on LoCoMo, where a cosine index recovers every annotated turn for only **22.7%** of multi-hop questions at $k = 20$ while recovering at least one for 80.1%. A second index of atomic facts, each citing the turn it came from, plus two small structural bonuses, raises the complete-evidence rate to **36.2%** and the any-evidence rate to 87.2% with nothing learned and under a millisecond per query. Every term is ablated; the fact index carries most of the gain, and two plausible additions—fusion of whole ranked lists and a surface-entity hop—make the result worse.

We then argue that this recall number is a unit test, not a product score. LoCoMo question answering is saturating under vendor protocols (Mem0 reports 92.5%, Zep 94.7%, on protocols that differ from each other), and the LoCoMo-Conv benchmark shows that the same memories phrased as a user would phrase them expose retrieval gaps the QA form hides. We describe the memory record and the read path Hanzo Context uses to close those gaps—immutable raw memory; atomic facts with canonical entities and validity intervals; query-conditioned planning; bounded typed expansion instead of global fusion; a hop whose query is rebuilt from the state the previous hop resolved; and deterministic resolution of superseded facts—and the protocol on which we evaluate it: a declared dev/test split, a frozen commit, LoCoMo, MemoryAgentBench FactConsolidation and LoCoMo-Conv, with retrieval and answering reported separately and evidence graded EXACT, SUPPORTED or ANSWERED.

1 The Claim

A memory system for an agent is asked one thing: given what was said before, find what bears on what is being asked now. The dominant implementation embeds every turn, embeds the question, and returns the nearest neighbours. That works when the answer resembles the question. It fails, quietly, when the answer is *connected* to the question by something a sentence embedding does not carry—the same person named two sessions apart, the date a plan was made and the

date it happened, the fact that one turn answers the turn before it.

This paper makes one technical claim and one methodological one.

The technical claim is that linkage is a separate index, not a better embedding. Extract short facts from each turn, keep the pointer from every fact back to the turn it came from, give each fact its entities and its time, and let a question reach a turn through a fact when it cannot reach it through the turn’s own wording. Section 7 measures this in isolation: on LoCoMo

multi-hop questions the complete-evidence rate at $k = 20$ moves from 22.7% to 36.2%, and the ablation in Table 3 attributes most of it to the fact index.

The methodological claim is that the number we just quoted is the wrong one to optimise. It measures whether every turn an annotator marked was retrieved. A system can answer correctly from a different turn that says the same thing; the LoCoMo-Conv authors call this *silent grounding* and ship extra annotations because strict gold-turn recall misses it [2]. Section 2 sets out what we measure instead and why, and Section 6 fixes the protocol before any number on the test split is read.

2 What Is Being Measured

Three quantities are routinely conflated under “LoCoMo score”.

Evidence recall. For each question LoCoMo lists the dialogue turns that support the answer [1]. **ALL@ k** is the fraction of questions for which every listed turn is in the top k ; **ANY@ k** the fraction for which at least one is. The two disagree by a factor of three on multi-hop questions (Table 1). **ALL** is the honest number for a retriever that must hand a reader complete evidence; **ANY** is the honest number for a retriever whose reader can follow a lead.

Answer accuracy. A reader is given the retrieved turns and scored on what it says: token-level F1 and exact match against the gold answer in the LoCoMo paper, or an LLM judge in most vendor reports. This is what Mem0’s 92.5% [9] and Zep’s 94.7% [11] are, and they are not the same measurement as each other: Zep’s page names gpt-5.4 with medium reasoning as reader and a chain-of-thought gpt-5.4 judge, median 5,760 context tokens per question, and 1,459 of 1,540 questions correct; Mem0’s post names an April 2026 algorithm, 6.7–7.0K tokens per retrieval, and states that Zep’s figure is disputed by a third-party re-run at 75.1%. Neither is comparable to the LoCoMo paper’s own F1 figures, and

neither is comparable to evidence recall.

Conflict resolution. MemoryAgentBench’s FactConsolidation task numbers its facts, later facts supersede earlier ones, and the question asks for the current value, one or two hops deep [3]. Its metric is **substring_exact_match**. CAR [5] reports 78.0 / 30.2 on single- and multi-hop with gpt-4o-mini and 94.8 / 51.5 with gpt-4o, pooled over context lengths from 6K to 262K tokens; at 262K the same pipelines score 82 / 27 and 93 / 41, against 54 / 5 for HippoRAG-v2 and 60 / 5 for gpt-4o reading the whole context. Pith’s page claims 68.0 exact match on the 262K multi-hop lane [20]. Naïve publishes 91 / 57 pooled on a matched gpt-4o-mini backbone [19]. That page names no dataset and gives no protocol, but it is identifiable as this benchmark from its own comparison column: its baselines are CAR’s to the decimal — 78.0 / 30.2, and 54, 28, 18 and 7 single-hop for HippoRAG-v2, MemGPT and Cogneer, Mem0 and Zep, whose multi-hop scores it collapses into a single “ $\leq 7\%$ ”. A table is identified by its baselines.

We report each quantity under its own name, never one as another.

Three grades of evidence. For a question with gold turns G and retrieved turns R , a system scores **EXACT** if $R \cap G \neq \emptyset$ (and **ALL** if $G \subseteq R$), **SUPPORTED** if some turn in R shares an atomic fact with a turn in G or contains the gold answer string, and **ANSWERED** if the reader’s answer is correct. The three are reported side by side. The gap between EXACT and SUPPORTED is the annotation’s blind spot; the gap between SUPPORTED and ANSWERED is the reader’s.

3 Related Work

Benchmarks. LoCoMo [1] provides ten very long two-person conversations with question–answer pairs and their supporting turns, in single-hop, multi-hop, temporal, open-domain and adversarial categories. LongMemEval [4]

tests extraction, multi-session reasoning, temporal reasoning, knowledge updates and abstention over scalable chat histories. MemoryAgentBench [3] recasts long-context tasks as incremental multi-turn interaction and adds FactConsolidation, the conflict-resolution task used here. LoCoMo-Conv [2] rewrites LoCoMo’s memories into dialog, implicit, counterfactual and composed requests—1,986, 1,986, 1,540 and 1,069 items—and standardises the memory systems it compares on all-MiniLM-L6-v2 embeddings, $k = 10$, gemma-4-31B-it at temperature 0 as reader, and GPT-5.4-mini with reasoning as judge, validated against Claude Sonnet 4.5 and Qwen3.6-35B-A3B on a 1,488-item subset. Without rewriting, Recall@10 falls from 0.531–0.659 on dialog to 0.308–0.456 on implicit queries across five systems.

Systems. MemGPT [18] manages context as an operating system manages memory tiers. A-MEM [17] organises notes Zettelkasten style. HippoRAG [16] runs personalised PageRank over an extracted graph. Mem0 [8] extracts and consolidates memories, and its 2026 algorithm adds single-pass add-only extraction and entity linking [9]. Zep [10] keeps a temporally aware knowledge graph and retrieves across facts, entities, episodes, observations and thread summaries [11]. AnchorMem [12] extracts atomic facts as retrieval anchors while preserving the raw context immutably, and binds related facts into event nodes rather than bridging through generic entities. AtomMem [13] selects high-value atomic facts and organises them into event structures and temporal profiles. Memora [14] indexes a short primary abstraction and several cue anchors per memory and keeps the full value unindexed, reporting up to 98% fewer context tokens. APEX-MEM [15] stores temporally grounded events in an entity-centric property graph with append-only history and resolves conflicts at query time, reporting 88.88% on LoCoMo QA and 86.2% on LongMemEval under its setup. Our design takes from each what their own ablations support: anchors over raw memory, events over entity hops, abstractions beside

cue anchors, append-only temporal history.

Conflict resolution. CAR [5] separates evidence extraction from policy: retrieve, extract numbered facts, pick the latest serial with code, then form the next hop. TEPA [6] makes validity an explicit, revocable state of a memory. MemTxn [7] gives memory a transaction boundary with source-supported admission and a temporal resolver. All three converge on the same rule: do not ask the reader which of two contradicting facts is current.

4 The Memory Record

Every write produces one immutable raw memory and a set of derived, indexed views of it. Nothing derived replaces the raw text; the reader is always handed the original.

- **raw** — the turn or document as received, with **source** (conversation, session, turn) and **observed_at**.
- **facts[]** — atomic statements, several per turn, each with a canonical **subject**, a **relation**, an **object** that is a canonical entity or a literal, an **event_time** when the text carries one, **valid_from** / **valid_to** where derivable, a **confidence**, an embedding, and the pointer to its source turn.
- **entities[]** — canonical identifiers resolved by type (person, place, organisation, work, date), not by capitalisation. The surface heuristic was measured and rejected (Section 13).
- **events[]** — membership of facts in a shared event, the associative link AnchorMem found to beat entity bridging.
- **supersedes[]** / **superseded_by[]** — a supersession key so that the current value of a relation is a lookup, not a judgement.
- **primary_abstraction** and **cue_anchors[]** — a short phrase saying what the memory is about and several context-aware tags

that give the same memory alternative entry points.

The fact layer is the linkage index. On LoCoMo the dataset’s own observation annotations cite 79.9% of multi-hop gold turns; the remaining 20% are unreachable through any fact, which bounds what a re-ranker can do and is why extraction coverage, measured in Table 6, comes before tuning.

5 The Read Path

5.1 Query understanding

The question is classified into retrieval operators—semantic, entity, relationship, temporal, current-value, causal, multi-hop, implicit—by rules over its surface (a year, a month, a name, “after”, “still”, “most recent”) with a small model consulted only where the rules are silent. The plan is an explicit state: what is resolved (an entity, a date, an event), what is unresolved (the relation or object being asked for), the evidence gathered so far, and the budget in hops, candidates, latency and context tokens. The weights of the generators below are conditioned on the plan; a temporal question spends its budget on the timeline, a current-value question on the supersession chain.

5.2 Seed retrieval and bounded expansion

Seeds come from dense search over turns, BM25 over turns, dense search over facts (each cited fact adds its source turn), exact matches on canonical entities, and the session that a dated question names. Each generator contributes a small pool—forty dense turns, twenty facts, three neighbours in the shipped configuration—rather than a long ranked list. Expansion is then bounded and typed: at most eight turns through event membership, at most eight through the timeline, only along edges whose confidence clears a threshold. A turn enters the pool because its text matched, because a fact it contains matched, or because a high-confidence typed link

from already-resolved evidence points to it. It does not enter because some generator ranked it forty-seventh.

5.3 One scorer

Every candidate is scored once. In the configuration that shipped in `hanzoai/memory`, the score is the turn’s cosine to the question, plus 0.2 times the cosine of the best fact citing it, plus 0.06 if it sits beside one of the top three turns in the same session, plus 0.05 if it sits in the month or year a dated question names. The trace records each term, so a rank is attributable to a mechanism.

Facets—three to five rewrites of an implicit or composed request—feed the same seed stage with the same small pools. LoCoMo-Conv reports that multi-facet rewriting lifts AnchorMem’s implicit Recall@10 from 0.368 to 0.524 while abstractive systems gain little [2]; our own earlier attempt to fuse whole ranked lists with reciprocal-rank fusion lost four points (Section 13), so facets are pooled small and scored once, never fused.

5.4 Stateful hops

The second hop is not a second embedding of the first sentence. When hop one resolves an entity and an event—“Sarah” to an identifier, “leaving Acme” to a dated departure—hop two is the structured query *subject = that identifier, relation = employed_by, time after that departure*, answered from the fact graph and the timeline. Similarity to the original wording no longer matters for the evidence hop two needs.

5.5 Deterministic resolution

For questions that ask for a current or versioned value, and only for those, the candidate facts are grouped by supersession key and the latest serial or the latest **valid_from** is selected in code before the reader sees anything. This is CAR’s rule [5] applied to an index that already carries the version structure CAR had to reconstruct at inference time. The reader is given the winning fact and its raw source turn; it is never asked to adjudicate between contradictions.

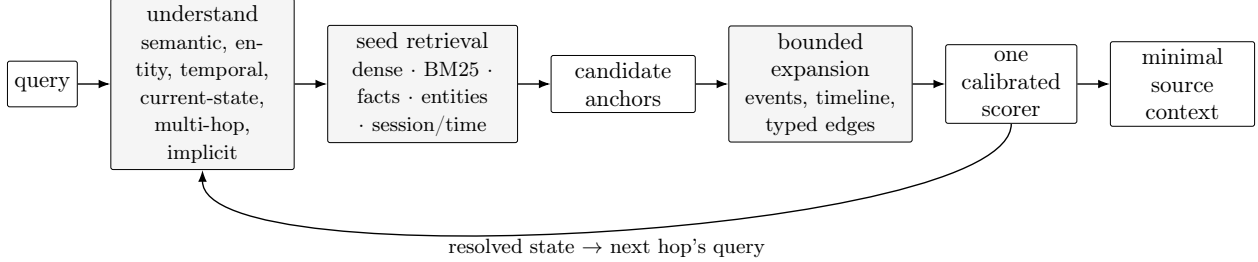


Figure 1: The read path. Every candidate enters through a named generator and carries the contribution of each term that scored it, so a ranking is explainable after the fact. A hop that resolves an entity or a date rewrites the query for the next hop; the original sentence is embedded once and never reused for hop two.

6 Protocol

The protocol is fixed in `METHOD.md` of the benchmark repository and is repeated here so that no number below can be read without it.

Splits. LoCoMo: dev is conversations 0–2, test is 3–9. The retrieval figures in Section 7 predate the split and were measured on all ten conversations; they are labelled “all” and are not re-quoted as test numbers. MemoryAgentBench FactConsolidation: dev is the two 6K haystacks (single- and multi-hop), test is 32K, 64K and 262K. LoCoMo-Conv: the paper’s protocol; dev is the first two conversations’ items, test the rest.

Metrics. Retrieval: R@5/10/20 ALL and ANY, MRR of the first gold turn, NDCG@20, candidates examined, context tokens delivered, p50 and p95 latency. Answering: token F1 and exact match on LoCoMo, substring exact match on MemoryAgentBench, the benchmark’s judge on LoCoMo-Conv. Bootstrap 95% intervals, 1,000 resamples over questions, on every headline.

Readers. Three lanes, never mixed in a table: *house*, enso-flash; *open*, google/gemma-4-31b-it, the reader LoCoMo-Conv standardises on, with qwen3-30b-a3b-instruct as fallback; *matched*, the reader a compared paper used where the router serves it. Temperature 0, 64 output tokens, one prompt file per benchmark shared by every row. Rows in one table share reader, prompt, *k* and context budget.

Freeze. A configuration is chosen on dev, its commit hash written beside its table, and the

test split run once. A later change is a new run identifier beside the old, never over it. No per-question routing, no tuning against test labels.

7 LoCoMo: Retrieval in Isolation

Ten conversations, 5,882 turns, 1,540 questions with evidence: 282 multi-hop, 321 temporal, 96 open-domain, 841 single-hop; the 446 adversarial questions are excluded, as Mem0 and Zep exclude them. Turns and questions are embedded with zen-embedding-0.6b; the dataset’s observation annotations serve as the fact layer for this experiment, each observation citing the turns it was extracted from. Nothing is learned. Two systems are measured. The first cut (Table 1) is the four-term pool and re-rank that shipped in `/v1/memory/search`: cosine over turns plus the best citing fact, adjacency and a time cue. The engine of Section 5 follows (Tables 3–5), with its weights and pools chosen on conversations 0–2 and frozen at commit 33d0f8749bd1 before conversations 3–9 were run once.

Where the misses are. For the gold turns cosine fails to retrieve, we measured which link would have reached them from a turn it did retrieve: adjacency 16.9%, same session 33.0%, a shared name 20.1%, a citing fact 23.9%, any of these 52.2%. Each link is weak alone and real in combination, and each only makes sense as a bonus on a candidate that already has a reason

R@20 (all)	single-hop		multi-hop	
	ALL	ANY	ALL	ANY
cosine over turns	78.2	80.7	22.7	80.1
linked	81.8	83.8	36.2	87.2

Table 1: Evidence recall at $k = 20$, in percent. The multi-hop complete-evidence rate rises by 13.5 points; the any-evidence rate by 7.1. Retrieval stays under a millisecond per question.

multi-hop ALL, first cut (all ten)	$k = 5$	$k = 10$	$k = 20$
	13.5	23.4	36.2
frozen configuration	$k = 5$	$k = 10$	$k = 20$
test split, multi-hop ALL	15.9	26.4	37.5
test split, multi-hop ANY	69.2	78.8	88.0
test split, single-hop ALL	74.1	81.7	87.7
all ten, multi-hop ALL	17.4	27.0	39.0
all ten, multi-hop ANY	68.8	79.8	87.9
all ten, single-hop ALL	72.7	80.4	86.8

Table 2: The same configuration by k . A number quoted from this table carries its k ; a reader that is handed ten turns sees 23.4%, not 36.2%.

to be there—which is the design in Figure 1 and the reason list fusion failed.

The ceiling. 79.9% of multi-hop gold turns are cited by at least one observation; for those, the best citing fact sits at median rank 8 in the fact index where the turn itself sits at median rank 20 in the turn index. The fact index is closer to the answer than the turn index is. The other 20.1% cannot be reached by any re-ranking of this fact layer, so the next section measures the fact layer itself.

8 Extraction Coverage

9 Parameter Search on the Dev Split

10 LoCoMo: Answers

A reader is handed the retrieved turns and scored on what it says (Table 8). Every row of a reader shares the prompt (prompts/reader-locomomo.txt), temperature 0 and a 64-token cap; rows differ only in what was retrieved. The open reader is gemma-4-31B-it, run locally, the reader LoCoMo-Conv fixes; qwen3.6-35B-A3B follows on all four categories. The oracle row hands the reader exactly the annotated turns.

With qwen3.6-35B-A3B over all 1,525 questions of the four categories, cosine context yields F1 48.8, the first cut 50.3, the engine 53.3 and the annotated turns 61.8; exact match 26.2, 26.7, 29.0 and 34.6. Retrieval moved the share of contexts holding a gold turn from 79.3% to 87.0%, and the engine’s rows are above cosine’s on every category (multi-hop 36.7 $\rightarrow$ 43.0, single-hop 56.9 $\rightarrow$ 61.9, temporal 47.5 $\rightarrow$ 49.5). The multi-hop questions with Gemma say where the remaining error is. Cosine context yields F1 37.6; the first cut 40.7; the engine 40.5; the annotated turns themselves 53.0. Retrieval moved the share of questions whose context holds a gold turn from 79.8% to 87.9%, and the reader turned that into three points of F1 out of a fifteen-point gap to the oracle. The rest of the gap is the reader’s: a 31B open model at temperature 0, asked for the shortest answer, scores 53.0 even when handed nothing but the evidence. That is the ceiling any retrieval improvement is measured against on this reader, and it is the reason the retrieval tables are reported on their own: a reader can hide a retrieval gain, and a stronger reader raises every row.

11 MemoryAgentBench: Fact-Consolidation

Eight haystacks: single- and multi-hop at 6K, 32K, 64K and 262K tokens, 100 questions each,

Configuration	multi-hop		single	temporal	open	all	nDCG	supported	pool	p50
	ALL@20	ANY@20	ALL@20	ALL@20	ALL@20	R@20			cands	ms
cosine over turns	22.7	79.8	78.2	77.3	33.7	71.9	49.6	81.0	40	0.55
+ BM25	22.7	78.0	82.5	78.8	37.0	74.6	53.2	83.3	52	0.61
+ atomic fact index	36.9	88.3	82.3	82.6	42.4	78.2	59.6	86.4	71	0.83
+ adjacency	39.0	88.7	84.2	83.5	44.6	79.7	61.4	87.8	74	0.83
+ second hop from resolved state	39.0	87.9	86.8	84.4	45.7	80.9	62.6	88.0	76	1.65
global reciprocal-rank fusion	33.7	89.0	86.8	83.8	40.2	80.5	56.2	89.0	137	1.73

Table 3: The engine on all ten conversations at $k = 20$, one generator added per row, the frozen weights throughout. ALL is the complete-evidence rate, ANY the any-evidence rate; R@20 is $|\text{retrieved} \cap \text{gold}|/|\text{gold}|$; *supported* counts a question whose top-20 holds a gold turn, a turn sharing an atomic fact with a gold turn, or a turn containing the gold answer. The fact index carries most of the gain; the second hop moves single-hop and temporal questions; a fusion of whole ranked lists loses nDCG for a little ANY.

Configuration	multi-hop		single	temporal	open	all	nDCG	supported	pool	p50
	ALL@20	ANY@20	ALL@20	ALL@20	ALL@20	R@20			cands	ms
cosine over turns	18.8	80.3	78.3	72.3	35.6	70.9	48.8	80.5	40	0.55
+ BM25	21.2	78.4	82.8	73.6	38.4	73.9	52.9	83.2	53	0.62
+ atomic fact index	35.1	88.5	83.0	78.8	43.8	77.9	58.7	86.4	71	1.15
+ canonical entities	35.1	88.5	83.0	78.8	43.8	77.9	58.7	86.4	71	1.10
+ timeline	34.6	88.5	83.3	79.7	45.2	78.2	59.9	86.8	71	1.04
+ adjacency	37.5	88.5	84.6	80.1	45.2	79.1	60.5	87.5	75	1.03
+ typed expansion	37.5	88.5	84.6	80.1	45.2	79.1	60.5	87.5	75	0.98
+ second hop from resolved state	37.5	88.0	87.7	81.4	45.2	80.7	61.6	87.8	77	1.67
pseudo-relevance feedback	19.7	75.5	72.4	69.3	35.6	66.4	46.3	75.8	634	1.10
chain search (top turn as query)	37.5	88.0	83.8	78.8	45.2	78.5	56.7	86.3	79	1.64
surface-entity expansion	35.6	88.0	83.0	78.8	43.8	77.9	58.2	86.2	82	0.98
global reciprocal-rank fusion	32.2	88.5	86.7	80.1	38.4	79.6	55.5	88.4	139	1.20

Table 4: The same rows on the test split (conversations 3–9), run once after the freeze. The failed shapes are below the rule: pseudo-relevance feedback drifts the query toward its own top three and examines 634 candidates to lose; chain search finds the multi-hop turns but scrambles the ranking; surface entities and global fusion are the two results that shaped the design.

facts numbered in order of arrival, later serials superseding earlier ones. The deterministic resolver of Section 5 is scoped to exactly this question type.

The write side parses every fact line into (serial, subject, relation, object) — 45 templates, no line left unparsed at any size, and the gold answer is the object of some fact for every question. The read side compiles the question into a base entity and a chain of relations, follows the chain hop by hop, and at each hop takes the latest serial for (entity, relation) deterministically; the reader, where one is used, sees only the resolved facts. The *noreader* row emits the resolved object string itself, so it is the resolver’s accuracy with no model in the loop at all.

Three resolvers stand in Table 9, each devel-

oped on the 6K haystacks and run on the others once. The first run compiled plans with a hand-written lexicon. It scored 98.0 / 77.5 (SH / MH) on the 6K dev haystacks, held single-hop at 92.9–98.0 on test, and fell to 32.0 / 31.3 / 25.5 on multi-hop at 32K / 64K / 262K. The traces say why: 126 of the 209 multi-hop misses were complete chains with the wrong answer, and 83 asked for a relation the entity never had. Both are the plan: the lexicon emitted relations in the question’s surface order (“capital of the country that X originates from” became [capital, country_origin] rather than the reverse) and added a hop for every relation word it saw, and it had been written against the 6K phrasings. Parse coverage, entity lookup and version selection were not the problem. The second run keeps

Configuration	multi-hop		single	temporal	open	all	nDCG	supported	pool	p50
	ALL@10	ANY@10	ALL@10	ALL@10	ALL@10	R@10			cands	ms
cosine over turns	9.6	49.6	51.4	46.1	16.3	45.3	31.1	53.1	40	0.25
+ BM25	13.8	58.9	67.2	62.0	23.9	59.2	43.7	67.6	54	0.31
+ atomic fact index	20.9	73.4	73.1	72.0	25.0	66.7	52.0	75.6	73	0.42
+ adjacency	21.6	74.5	74.6	72.6	27.2	67.7	53.3	76.3	77	0.42
+ second hop from resolved state	24.1	78.0	77.5	73.5	27.2	70.1	55.8	78.5	78	0.76
global reciprocal-rank fusion	20.2	74.8	73.4	74.1	28.3	67.2	48.6	76.1	143	0.43

Table 5: The same engine over all-MiniLM-L6-v2 vectors at $k = 10$, the embedder and cut LoCoMo-Conv fixes for every system it compares, all ten conversations. With a weak embedder linkage matters more, not less: the complete-evidence rate on multi-hop questions rises from 9.6 to 24.1 and nDCG from 31.1 to 55.8, from the same fact layer and the same weights.

Fact layer	gold turns with ≥ 1 fact	with the right entity	with a dated fact
LoCoMo observations (dataset)	79.9% (multi-hop)	...	...
Hanzo extraction	...	...	...

Table 6: Fact coverage of LoCoMo gold turns. A gold turn is covered when at least one extracted fact cites it; covered with the right entity when that fact’s canonical subject or object is the entity the question names; covered with a dated fact when the fact carries an event time. Precision is read on a hand-checked sample of 100 facts.

the exact single-hop templates as rules, has a model compile every other question into the same plan shape, validated against the relation set, and uses the lexicon only where the model’s plan fails validation. On the shared GPU every model call of that pass timed out and the lexicon filled every plan, so its plans were the lexicon’s; it scored 59 / 59 / 33 on multi-hop at 32K / 64K / 262K, the gain over the first run being the fixed plan policy for single-hop and a corrected lexicon, not a model.

The third row treats the plan as a sketch. An oracle search over the parsed graph (`mab/oracle.mjs`) found a path from the plan’s base entity to the gold for 65 of the 67 multi-hop misses at 262K; the compiler’s errors were near-synonym relations (`language` for `official_lang`, `origin` for `country_origin`), chains cut a hop short or padded with one, and the order of two hops. A second measurement (`mab/versions.mjs`) settled, on dev, what “current” means: for single-hop questions the gold is the last version of its key 71 times and the first 0; for multi-hop chains 63 are reachable with the latest version at every hop and none with the first. So the search keeps “latest wins” and relaxes the plan:

at each hop the planned relation is preferred, its class-mates allowed at a cost, an older version taken only when the latest leads nowhere, one typed bridge through the entity’s own facts when it has no such relation, a planned relation nothing satisfies skipped at a cost, and the answer type the question names — a city, a country, a person, a language — as the final constraint, deciding between stopping a hop early and going one further. The best complete path wins by a fixed score; ties go to the more literal plan and the later version. It is deterministic, it reads no gold, and it calls no model at execution time.

The plans it searches over are several per question: the lexicon’s, and those of Qwen3.8-27B (4-bit, local) compiled once at temperature 0 and once at 0.7, at the relation level and at the level of relation families, each also in every order of its relations. Every plan of every set is executed and the same score selects among them; `meta.json` records the planner of every run and the frozen archive carries every plan set with its sha256. The compiler is the one place a language model takes part, before any haystack is read; it never sees a fact. Table 10 separates the two: plan recall@N (some plan

Parameter	range searched	start	chosen
fact weight	0.10–0.40	0.2	0.3
BM25 weight	0–0.30	0.15	0.2
entity weight	0–0.20	0.1	0.1
temporal weight	0–0.12	0.05	0.12
adjacency weight	0–0.10	0.06	0.1
expansion weight	0–0.15	0.08	0.08
second-hop weight	0–0.45	0.3	0.3
facet weight	0–0.30	0.2	0.2
dense pool	20–60	40	40
fact pool	10–30	20	30
BM25 pool	10–30	20	20
second-hop pool	5–15	10	15

dev objective (mean over categories of $(\text{ALL@20} + \text{nDCG@20})/2$): $61.3 \rightarrow 63.7$ over 145 evaluations
frozen at commit `33d0f8749bd1`; test split, run once: multi-hop ALL@20 37.5, ANY@20 88.0, single-hop ALL@20 87.7

Table 7: Every parameter searched, the range, the value chosen on dev, and the test number read once after the freeze. The full grid is in [ablations/](#).

among the first N sets reaches the gold) against execution success@N (the selected one does). Recall climbs with every set and selection tracks it within a point or two — 68 against 67 at 262K — so the sets, not the selector, set the number. A third row asks a language model to repair a plan the selector finds suspicious, and it is in the table at its dev numbers and nowhere else: across 196 dev questions the repair fired *once*, that one plan was chosen, and it is the whole of the row’s 88 to 89. One question, for a 27B model held resident beside the search. It was not promoted to test, which is why its test cells are empty — a configuration that loses on dev does not get a second chance on held-out data, and reporting the empty cells is cheaper than pretending the row was never run. The fourth row adds an exhaustive enumeration of the relaxed plan and a lexical tie-break from the compiler’s own templates, and buys five points where the repair bought one, with no model in the loop at all. On dev it moved multi-hop from 88 to 93 and single-hop from 98 to 100; on test, run once, multi-hop 89 / 89 / 67 and single-hop 98 / 98 / 97. Pooled, 98.3 / 84.5, counting every unanswered question as wrong.

Against the reference points: CAR’s gpt-4o pipeline reports 94.8 / 51.5 pooled and 93 / 41 at 262K; this search reports 98.3 / 84.5 pooled and 97 / 67 at 262K with no reader. Naïve’s 91

/ 57 is pooled on the same benchmark, identified as above, on a matched gpt-4o-mini backbone; this search reports 98.3 / 84.5 with no reader at all, +7.3 single-hop and +27.5 multi-hop. The per-lane split is not published there, so the comparison is of pooled figures, and what varied under their measurement is unstated. Pith reports multi-hop 95 / 83 / 84 / 68 at 6K / 32K / 64K / 262K under its own protocol [20], a four-lane mean of 82.5 against this search’s 84.5: above it at 32K and 64K, two below at 6K and one below at 262K, inside the bootstrap interval on every lane. Its protocol is undisclosed, so this compares reported numbers, not systems, and we claim no best published result on the benchmark.

What stops 262K at two thirds. The search’s 33 misses at 262K are 29 questions whose gold uses an *older* version of a key that a later line updates, and 4 where a shortcut path hides the same choice; every one of the 100 has a gold path in the parsed store (Table 11). The later line is an edit no question in the haystack asks for: the benchmark is built from MQuAKE-CF counterfactual edits, a haystack packs the edits of instances it never asks about, and the gold follows the edit on 183 versioned hops and the original on 46. Two ceilings follow. A key is contested when two questions need it in different versions; a policy choosing one version per key,

Context	Reader	Questions	n	F1	EM	EXACT	ANSWERED	tokens/q	reader p50 s	95% CI (F1)
engine @k=20	enso-flash	multi-hop	282	45.1	13.8	87.9	47.5	971	5.0	[41.2, 49.1]
oracle gold	enso-flash	multi-hop	282	53.6	17.7	99.6	60.6	169	7.0	[49.2, 57.8]
cosine @k=20	gemma4:31b	multi-hop	282	37.6	9.9	79.8	42.6	835	16.1	[33.6, 41.5]
first cut @k=20	gemma4:31b	multi-hop	282	40.7	11.0	87.2	44.7	968	19.5	[36.7, 44.9]
engine @k=20	gemma4:31b	multi-hop	282	40.5	9.9	87.9	44.0	971	18.1	[36.4, 44.3]
oracle gold	gemma4:31b	multi-hop	282	53.0	16.0	99.6	58.5	169	4.8	[49.0, 57.0]
engine @k=20	gpt-oss-120b	multi-hop	282	45.3	13.5	87.9	47.9	971	2.7	[41.6, 49.5]
oracle gold	gpt-oss-120b	multi-hop	282	55.5	18.1	99.6	61.3	169	2.4	[51.4, 59.5]
cosine @k=20	qwen3.6:35b-a3b	all four	1525	48.8	26.2	79.3	51.3	841	4.6	[46.6, 51.0]
first cut @k=20	qwen3.6:35b-a3b	all four	1525	50.3	26.7	84.1	52.9	970	3.4	[48.2, 52.4]
engine @k=10	qwen3.6:35b-a3b	all four	1536	51.1	27.9	81.1	54.1	501	1.5	[48.9, 53.2]
engine @k=20	qwen3.6:35b-a3b	all four	1525	53.3	29.0	87.0	55.9	969	5.0	[51.2, 55.4]
engine @k=5	qwen3.6:35b-a3b	all four	1536	47.6	25.5	73.8	50.3	249	1.1	[45.5, 49.7]
oracle gold	qwen3.6:35b-a3b	all four	1525	61.8	34.6	99.7	66.0	82	2.9	[59.7, 63.8]

Table 8: Answer accuracy on LoCoMo with the same reader, prompt and k across rows, all ten conversations. EXACT is the share of questions whose context held an annotated turn; ANSWERED is $F1 \geq 0.5$ or an exact match. The oracle row hands the reader exactly the annotated turns and bounds what retrieval can still buy. Tokens per question are the retrieved context delivered to the reader; the reader’s own p50 latency is the local Ollama service under shared load and is reported, not claimed.

however it chooses, can satisfy one side of each contest, which caps an oracle at 94 at 262K. “Latest wins” caps at 67, and the search is on that line. Six signals a store can read were tried against the 46 older-gold hops, with the 183 latest-gold hops as control: the object’s degree (gold higher 21, lower 20), whether the object has facts of its own (equal), whether it is tied back to its subject elsewhere (13 to 0 for the gold — and 4 to 42 on the control, because the tie marks the *original* fact, which is the gold one time in five), contest by another question (6 questions), serial distance to the chain’s other facts (24 to 38), and position in the list by decile (one distribution for followed and ignored edits). None separates them. A system that reads only the haystack cannot pass 67 at 262K without guessing; the published 68 is one question above that line. The residue is a property of the benchmark at that size, and the way past it is not a reader but the question’s own instance — which the haystack does not carry.

12 LoCoMo-Conv

The benchmark’s own protocol: all-MiniLM-L6-v2 embeddings, $k = 10$, $\text{recall} = |\text{retrieved} \cap \text{gold}|/|\text{gold}|$ with a gold turn counted when its verbatim text appears in a returned memory,

gemma-4-31B-it at temperature 0, GPT-5.4-mini as judge. Reference points from the paper [2]: without rewriting, Recall@10 on dialog queries spans 0.531–0.659 across the five systems, with AnchorMem at 0.659; on implicit queries 0.308–0.456, with mem0 at 0.456 and Memora at 0.445; multi-facet rewriting lifts AnchorMem’s implicit recall from 0.368 to 0.524 and its composed recall from 0.279 to 0.432.

At the time of writing the data is not released: the paper’s one release address, github.com/MiuLab/LoCoMo-Conv answers 404, and no copy exists on Hugging Face or in the authors’ other repositories. We therefore report no LoCoMo-Conv number. Two things could be done without the data, and were. First, the harness: `conv/retrieve.mjs` implements the protocol above against any ranking function, with the four styles as a loader slot, so the run is one command when the data appears. Second, a sanity check of the protocol on the questions the rewrites came from (Table 13): the same embedder, the same k , the same containment rule, on the original LoCoMo questions. Single turns as the memory unit reproduce none of the paper’s rows; two-turn exchanges land within three points of its Naive RAG dialog row. The paper does not state its memory unit, and its containment rule pays for larger units, which is worth knowing before any system’s number on

Configuration (reader)	FC-SH					FC-MH				
	6k	32k	64k	262k	pooled	6k	32k	64k	262k	pooled
resolver, latest wins (no reader)	96.0	97.0	95.0	94.0	95.5	76.0	59.0	59.0	33.0	56.8
typed search over the plans (no reader)	98.0	97.0	97.0	94.0	96.5	88.0	87.0	84.0	64.0	80.8
typed search, exhaustive, lexical tie-break (no reader)	100.0	98.0	98.0	97.0	98.3	93.0	89.0	89.0	67.0	84.5
typed search with one repair call (no reader)	98.0	...	...	...	...	89.0	...	...	...	...
CAR (arXiv 2606.01435), pooled 6k–262k: 78.0 / 30.2 with gpt-4o-mini, 94.8 / 51.5 with gpt-4o; at 262k: 82 / 27 and 93 / 41										
Naïve Brain, usenaive.ai/lab/memory (read 2026-09-15, dated July 2026), pooled: 91 / 57 on a matched gpt-4o-mini backbone										
that page names no dataset; it is identified as this one by its baselines, which are CAR’s to the decimal — 78.0/30.2, 54, 28, 18, 7										

Table 9: Substring exact match on FactConsolidation. Rows name their reader; *none* is the resolver’s own output. The 6K columns are the dev split; 32K–262K were run once per configuration. Pooled is the mean of the four sizes, as CAR reports its headline.

Haystack	sets	plans / question	N=1	N=3	N=9		6k	32k	64k
6k	9	13.8	88 / 88	90 / 90	94 / 94	multi-hop questions with a gold path in the parsed store	96	98	99
32k	9	14.6	79 / 79	84 / 81	89 / 89	gold needs an older version of an updated key	3	13	13
64k	9	16.6	83 / 83	86 / 85	88 / 88	keys contested between questions	4	6	6
262k	9	15.3	53 / 53	62 / 59	68 / 67	ceiling, one version per key chosen by an oracle	97	94	99
						ceiling, latest version wins	97	87	99
						the search, selected-of-N	93	89	88

Table 10: Best-of-N against selected-of-N on the multi-hop haystacks: plan recall@N / execution success@N out of 100, no reader. Sets enter in the lane’s order; every set is also executed in every order of its relations.

it is compared with another’s. The same check with zen-embedding-0.6b in place of MiniLM moves single-turn recall by 17 points before any linkage, which is why Table 5 holds the embedder fixed.

13 Negative Results

Two additions that read as improvements were measured and removed.

Global reciprocal-rank fusion. Fusing the turn ranking, the fact-cited ranking and the neighbour ranking as whole lists lost four points of multi-hop ALL@20 against the pooled re-rank. Every weak nominator pushed sixty candidates in; the one turn the fact index was meant to lift from rank 67 to rank 12 was drowned by fifty-nine others it lifted for no reason.

Table 11: The ceiling of a consistent version policy per multi-hop haystack, out of 100. A contested key is one that two questions need in different versions; the oracle row picks one version per key to satisfy the most questions, the latest-wins row is the policy every row of Table 9 uses.

Surface-entity expansion. Treating capitalised words shared between the nearest facts as entities and hopping through them lost two points. “Sunday” is not an entity. Entities must be resolved by type, and expansion must be bounded and typed, which is the design of Section 5.

Pending. Pseudo-relevance feedback, generic multi-query expansion and chain search are run as rows in the ablations of Tables 9 and 13; they are reported whether or not they help.

14 Hanzo ContextBench

External benchmarks are run unchanged. Beside them we are building a benchmark that measures what a company brain is for, with ten dimen-

System, 262K	reader	FC-SH	FC-MH
CAR pipeline [5]	gpt-4o-mini	82	27
CAR pipeline [5]	gpt-4o	93	41
gpt-4o, whole context [5]	gpt-4o	60	5
HippoRAG-v2 [5]	—	54	5
BM25 [5]	—	48	3
Cognee, MemGPT [5]	—	28	3
Mem0 [5]	—	18	2
Zep/Graphiti [5]	—	7	3
Pith (vendor page) [20]	undisclosed	—	68.0

Table 12: Published reference points at 262K, as reported by their sources. CAR’s pooled figures over all four lengths are 78.0 / 30.2 with gpt-4o-mini and 94.8 / 51.5 with gpt-4o. gpt-4o-mini is not served on our router, so the matched lane is reported only where a comparable reader could be run.

Memory unit, all-MiniLM-L6-v2, k=10	R@10 all (1,977)	no adversarial (1,531)	tokens/q
single turn	42.8	45.4	279
2-turn exchange	56.3	54.3	571
3-turn window	62.5	60.7	829
5-turn window	60.3	58.2	1258
single turn, zen-embedding-0.6b	59.7	63.5	315
paper’s Naive RAG row: dialog 0.533, implicit 0.312, counterfactual 0.573, composed 0.266			

Table 13: LoCoMo-Conv’s retrieval protocol on the original LoCoMo questions, the closest public proxy while the rewrites are unreleased. R@10 by memory unit, all-MiniLM-L6-v2. The paper’s Naive RAG row is given for scale; it is not the same question set and is not compared.

sions scored per item: answer, grounding, evidence completeness, contextuality, composition, temporal choice, conflict suppression, tokens retrieved, retrieval latency, and provenance—whether every claim in the answer traces to a source. Items are real conversational memory needs: explicit, implicit, temporal, conflicting, composed, causal, cross-session, cross-document, user and organisation scope, and memory combined with the current context. Evidence is graded EXACT, SUPPORTED or ANSWERED as in Section 2, so a system that answers from an equivalent turn is not scored as a miss. The same structure is applied to code in the companion paper: symbol, definition, reference, implementation, caller, callee, type, file, commit, issue, test, runtime log. We note that [21] independently uses the name ContextBench for a coding-agent retrieval benchmark; ours is named for the product it measures, and the two are

distinct.

15 Limitations

The fact layer used for the retrieval result in Section 7 is LoCoMo’s own annotation, extracted by the dataset authors with a strong model; Section 8 replaces it with our extraction and reports coverage and precision so that the substitution is visible. The retrieval result was measured on all ten conversations before the split was declared and is therefore not a test number. Two readers our comparisons name, gpt-4o-mini and gpt-5.4, are not served on our router; where a matched lane is missing the table says so rather than substituting. LoCoMo has ten conversations; intervals are wide and we print them. Vendor figures quoted in Section 2 are their own claims on their own protocols and are labelled as such.

16 Conclusion

Similarity finds resemblance. Context finds relationships. The measured part of this paper shows that a second index of atomic facts, each pointing at its source turn, lifts the complete-evidence rate on multi-hop questions by more than half with no learning and no latency, and that two natural extensions make it worse. The rest sets out a memory record and a read path built from what the field’s own ablations support—preserve everything, index several views, link typed facts and events, resolve time in code, change the query after every hop, return the least original context that proves the answer—and a protocol under which its numbers can be believed.

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